Strategic Data Source Assessment for the Fessi Knowledge Graph: Food & Grocery Waste Management

1. Architectural Alignment and Data Ontology Analysis

The development of "Fessi," a retrieval-augmented generation (RAG) system designed to navigate the complexities of German university waste management, necessitates a rigorous data acquisition strategy that transcends simple database lookups. The core challenge in building a knowledge graph for waste disposal lies in resolving the "Semantic Gap" between commercial product identification and environmental material science. Consumers identify objects by brand and name (e.g., "Haribo Goldbären"), while waste management infrastructure operates on material properties and legal classifications (e.g., "Polypropylene," "Composite," "AVV 15 01 02").

To successfully populate the three layers of the Fessi graph—Core Entities, Logic & Rules, and Infrastructure—we must integrate heterogeneous data sources that speak these different languages. The research conducted has identified a stratified ecosystem of data providers ranging from crowdsourced global repositories to strictly regulated German government mandates. The following analysis details the architectural fit of these sources, evaluating their capacity to feed the Graph-RAG system with high-fidelity structured data.

1.1 The Granularity Gap and Graph Schema Requirements

A critical insight emerging from the source analysis is the existence of a "Granularity Gap." Most standard product databases, such as standard GS1 lookups or retailer APIs, stop at the logistical unit level—providing weight, dimensions, and existence. However, precise waste disposal requires component-level granularity. A single product, such as a yogurt pot, represents a multi-node subgraph in the Fessi ontology: it consists of a plastic cup (Yellow Bin), an aluminum lid (Yellow Bin, but separated), and a cardboard sleeve (Paper Bin).

Therefore, the selected data sources must support an ontology that decomposes a single GTIN (Global Trade Item Number) into multiple PackagingComponent nodes. The research indicates that **Open Food Facts** is currently the only scalable, open-source dataset that attempts this level of decomposition via its packaging_tags taxonomy.[1] This sets the baseline requirement for the "Core Entities" layer: sources must support or allow for the derivation of 1:N relationships between products and materials.

1.2 The Logic Layer: From Static Rules to Dynamic Decision Trees

The "Logic & Rules" layer of Fessi cannot be populated by static lookup tables alone due to

the conditional nature of waste disposal rules in Germany. The "Mindeststandard" (Minimum Standard) published by the **Zentrale Stelle Verpackungsregister (ZSVR)** provides the ground truth for these rules.[3] However, this data exists largely in unstructured regulatory documents. The integration strategy must therefore involve a "Human-in-the-Loop" knowledge engineering phase where these regulations are translated into SHACL (Shapes Constraint Language) or similar rule languages within the graph.

For example, the ZSVR dictates that "highly soiled paper" moves from AVV 15 01 01 (Paper Packaging) to AVV 20 03 01 (Residual Waste). The data sources selected for the infrastructure layer, specifically the **Berliner Stadtreinigung (BSR) Abfall-ABC**, provide the semantic bridge for these rules, offering structured text that explicitly links material states (e.g., "stained," "empty") to disposal outcomes.[5]

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2. Primary Entity Source: Open Food Facts (OFF)

Open Food Facts represents the most significant available dataset for bridging the gap between consumer barcodes and packaging materials. Unlike commercial aggregators that focus purely on nutritional values for health applications or logistical data for supply chains, OFF has cultivated a dedicated, evolving taxonomy specifically for environmental and packaging data. It serves as the foundational "Entity Layer" source for Fessi.

2.1 Source Identification

- **Name:** Open Food Facts (World / Germany)
- **URL:** https://world.openfoodfacts.org/data
- **Provider:** Open Food Facts (Non-profit Association, France)
- **Type:** Crowdsourced Open Data (Wiki-style)

2.2 Data Format & Access

The accessibility of Open Food Facts is a major advantage for academic and non-profit projects like Fessi. The platform embraces a radical open-data philosophy, offering multiple ingress points for developers.

- **Format:**
  - **JSON API:** The primary interface for real-time querying. It returns deeply nested JSON objects containing product metadata, packaging hierarchies, and nutritional info.
  - **MongoDB Dump:** A daily snapshot of the entire database. This is critical for the initial "bulk load" of the Fessi graph, allowing for offline processing and sanitization of millions of records without hitting API rate limits.[6]
  - **JSONL / CSV:** Lighter-weight exports are available, though they often flatten the nested packaging structures required for the graph, making the MongoDB dump or

JSONL preferred for graph construction.

- **Access Method:**
  - **Free (Open Data):** No registration is required for read operations or data dumps.
  - **Authenticated Write:** Integrating write-back capabilities (e.g., if Fessi users correct a bin assignment) requires a free user account.
- **Rate Limits:**
  - **Read:** The API enforces a limit of approximately 100 requests per minute per IP address for product queries.[2]
  - **Search:** Search queries are more expensive and limited to 10 requests per minute.
  - **Bulk Policy:** The provider explicitly requests that high-volume users download the daily dump rather than crawling the API, which aligns with the Fessi architecture of building a local graph cache.[2]

2.3 Coverage & Quality Analysis

The utility of OFF for a German university chatbot depends heavily on its coverage of the DACH (Germany, Austria, Switzerland) region.

- **Record Count:** The database holds over 3 million products globally. Analysis of the countries_tags field indicates substantial coverage in Germany, with estimates exceeding 800,000 records relevant to the German market.[7]
- **Geographic Scope:** While global, the dataset allows for precise filtering using tags such as countries_tags:en:germany. This ensures the Fessi graph is not polluted with US-specific or Asian-specific packaging data that might not align with EU regulations.
- **Data Quality:** As a crowdsourced platform, quality is variable.
  - *Strengths:* High availability of barcode-to-name mappings.
  - *Weaknesses:* The packaging field was historically a free-text string (e.g., "cardboard box, plastic film"), which is difficult to parse programmatically. However, a massive migration is underway to the structured packaging_tags taxonomy (e.g., en:bottle, material:polyethylene-terephthalate).[1]
  - *Mitigation:* Fessi must implement a "Trust Score" for nodes, prioritizing data verified by the "Robotoff" AI system or trusted contributors over anonymous inputs.[9]

2.4 Licensing and Legal Implications

- **License Type:** Open Database License (ODbL) v1.0.[10]
- **Commercial Use:** Allowed, provided the derived database remains open if publicly distributed.
- **Attribution Required:** Yes. The chatbot must explicitly cite Open Food Facts when serving data derived from it.
- **Implications for Fessi:** Since Fessi is a query engine (chatbot) rather than a redistributed database product, the "Share-Alike" provision of ODbL allows for internal graph construction. However, if the university publishes the Fessi Knowledge Graph as a

standalone dataset, it must be licensed under ODbL.

2.5 Relevance Assessment (Score: 5/5)

- **Coverage Relevance (5/5):** It is the definitive open source for grocery GTINs.
- **Data Richness (4/5):** While specific material density data is sometimes missing, the explicit packaging_tags field is unique among free sources.
- **Accessibility (5/5):** Excellent documentation, zero cost, and multiple export formats make integration seamless.

2.6 Integration Notes & Technical Analysis

Critical Fields for Graph Mapping:
The ingestion pipeline for Fessi should focus on extracting specific JSON paths that map directly to the WasteItem and PackagingComponent entities.

OFF Field (JSON Path)	Fessi Graph Entity	Description / Usage
code	WasteItem.gtin	Primary Key (Barcode).
product_name_de	WasteItem.name	The human-readable string displayed to the user.
packaging_tags	PackagingComponent.material	The raw taxonomy tags (e.g., en:pet, de:glas).
ecoscore_grade	WasteItem.ecoScore	Can be used to provide "Green Nudges" in the chat.
ecoscore_data.adjustments.packaging	PackagingComponent Attributes	Contains nested data on shape, material, and recycling instructions derived from the tags.[11]
brands_tags	WasteItem.brand	Useful for entity resolution and disambiguation.

Taxonomy Normalization:
The OFF taxonomy uses a prefix system (en:, de:, fr:) that requires normalization. The Fessi ETL pipeline must strip these prefixes and map them to a controlled vocabulary based on AVV codes.

- **Source:** en:plastic-bottle -> **Fessi:** Material:Plastic + Shape:Bottle.
- **Source:** de:grüner-punkt -> **Fessi:** RecyclingScheme:DualSystem.

Example API Query:
To fetch a product's packaging data specifically:

```
HTTP

GET https://world.openfoodfacts.org/api/v2/product/40111410.json?fields=code,product_name,packaging_tags,ecoscore_data
```

The response includes the packaging_tags array, which might contain ["en:plastic", "en:tub", "en:aluminum", "en:lid"]. This allows the graph to construct a composite object requiring separation instructions.[12]

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3. Regulatory Data Source: ZSVR & Dual Systems

While Open Food Facts provides the *items*, the **Zentrale Stelle Verpackungsregister (ZSVR)** provides the *rules*. In the German waste management ecosystem, the ZSVR is the central authority mandated by the Packaging Act (VerpackG) to oversee the registration and recyclability of packaging. This data is essential for the "Logic Layer" of Fessi, providing the ground truth against which product materials must be evaluated.

3.1 Source Identification

- **Name:** ZSVR Mindeststandard Recyclingfähigkeit (Minimum Standard for Assessment of Recyclability)
- **URL:** https://www.verpackungsregister.org
- **Provider:** Stiftung Zentrale Stelle Verpackungsregister (Government-mandated authority) in agreement with the Umweltbundesamt (UBA).

3.2 Data Format & Access

The primary challenge with ZSVR data is its format. Unlike the JSON-native world of OFF, regulatory data is locked in document formats.

- **Format:** PDF Documents. The "Mindeststandard" is updated annually (e.g., "Mindeststandard 2024").
- **Access Method:** Free download from the ZSVR website.
- **Machine Readability:** Low. The data consists of technical tables, flowcharts, and

exclusion lists within PDFs.[3]
- **Bulk Download:** Not applicable in a database sense; requires downloading the regulatory documents.

3.3 Coverage & Quality

- **Scope:** Germany-wide. This is the federal legal standard that overrides local municipal variations regarding *what* is technically recyclable.
- **Granularity:** The data is extremely granular regarding material science. It defines specific combinations (e.g., "Paper packaging with >5% non-paper content") and their sorting compatibility.
- **Quality:** Authoritative. This is the legal basis for the Dual System (Yellow Bin/Gelber Sack).

3.4 Integration Notes: The "Human-in-the-Loop" Pipeline

Since this data cannot be ingested via API, the integration strategy for Fessi involves a manual knowledge engineering phase.

- **Rule Extraction:** A domain expert must translate the PDF regulations into the Fessi DisposalRule logic nodes.
- **Key Logic Mappings:**
  - **Composite Thresholds:** The Mindeststandard defines that fiber-based packaging (paper) with more than 5% non-paper material (barriers, coatings) that cannot be separated is often classified differently or poses recycling challenges. The Fessi graph must contain a rule: IF Material=Paper AND Coating=Plastic AND Separable=False THEN Stream=YellowBin (typically, though rules vary for pure composites).
  - **Black Plastics:** Historical sorting machines could not detect black plastic colored with carbon black. The Mindeststandard updates on this (e.g., detecting new NIR-detectable black pigments) must be reflected in the graph to give accurate advice.[14]
  - **LUCID Register:** The public LUCID register lists all producers. While it doesn't provide product-level data, it allows Fessi to validate brands found in OFF. If a brand in OFF is not in LUCID, the packaging might be non-compliant or grey-market, triggering a flag in the system.[15]

3.5 Relevance Assessment (Score: 5/5)

- **Coverage Relevance:** Essential. Without this, the chatbot cannot accurately determine recyclability claims.
- **Data Richness:** High density of logic, low density of structured entities.
- **Accessibility:** Low (requires manual digitization).

4. Municipal Data Source: Berliner Stadtreinigung (BSR)

Waste management in Germany is locally executed. While the ZSVR sets the federal standard, the local municipal provider determines the physical infrastructure (bins). For a German university, typically located in a city, the local "Abfall-ABC" (Waste ABC) is the dictionary that maps generic items to physical bins. **Berliner Stadtreinigung (BSR)** is highlighted here as a "Gold Standard" source because they publish their Abfall-ABC as structured Open Data, serving as a transferable ontology for other regions.

4.1 Source Identification

- **Name:** BSR Recyclinghöfe und Abfall-ABC
- **URL:** https://daten.berlin.de/datensaetze/bsr-recyclingh-fe-und-abfall-abc
- **Provider:** Berliner Stadtreinigung (BSR) via Berlin Open Data Portal.

4.2 Data Format & Access

- **Format:** JSON, CSV, XML.
- **Access Method:** Free download via the Berlin Open Data Portal.
- **License:** Creative Commons Attribution (CC BY).[5]
- **Update Frequency:** Irregular (as waste rules change), but generally static enough for annual updates.

4.3 Coverage & Quality

- **Record Count:** The dataset contains hundreds of common waste fractions (e.g., "Pizza box," "Mirror," "Lightbulb," "Yogurt cup").
- **Scope:** Berlin context. However, the logic for standard packaging (Yellow Bin vs. Paper Bin vs. Glass) is 95% transferable to other German cities due to the standardized Dual System.
- **Exceptions:** One key regional difference is the "Wertstofftonne" (Recyclables Bin) vs. "Gelber Sack" (Yellow Sack). Berlin uses the Wertstofftonne, which accepts non-packaging plastics. Fessi must have a configuration toggle for the specific university's location to adjust this logic.
- **Quality:** High. This is official municipal data used by the city's own apps.

4.4 Integration Notes

This dataset is perfect for the **"Concept Mapping"** phase of the Fessi graph.

- **Ontology Construction:** BSR data can be used to build the class hierarchy of the graph.
    - *Class:* WasteCategory (from BSR "Stoffarten").

    - *Property:* disposalInstruction (from BSR "Entsorgungstipp").
    - *Property:* avoidanceTip (from BSR "Vermeidungstipp").
- **Mapping Example:**
    - BSR Entry: "Aktenordner" (File folder) $\rightarrow$ Disposal: "Restmüll" (Residual).
    - BSR Entry: "Joghurtbecher" (Yogurt cup) $\rightarrow$ Disposal: "Wertstofftonne" (Recyclables).
- **JSON Structure:** The data includes fields like definition, storage, and disposal_costs, which add rich context to the chatbot's answers.[5]

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5. Verification & Enrichment Sources

To ensure high trust and detailed environmental data, the Fessi graph should incorporate secondary sources that validate or enrich the primary OFF and ZSVR data.

5.1 The "Trust" Layer: GS1 Germany

- **Source:** GS1 GEPIR / Verified by GS1.
- **Role:** GS1 is the issuer of GTINs. Their database is the source of truth for "Who manufactured this?"
- **Data:** Links GTIN to Company/Brand Owner.
- **Limitations:** The public GEPIR tool does not typically expose consumer-facing disposal instructions or detailed packaging materials. Access to the full "GS1 Data Hub" usually requires a paid license or membership.
- **Relevance (Score: 3/5):** Critical for **Entity Resolution** (verifying that a barcode exists and belongs to a specific company) but low for packaging material discovery.[17]

5.2 The "Sustainability" Layer: Blue Angel (Blauer Engel)

- **Source:** Blue Angel E-Commerce Data Service.
- **URL:** https://www.blauer-engel.de/en/trade/e-commerce
- **Role:** The world's oldest eco-label, covering categories like recycled paper, office supplies, and cleaning agents.
- **Data Access:** Offers a free data service for online retailers (and by extension, valid research projects) providing CSV/JSON exports of certified products.
- **Fields:** GTIN, Product Name, License Validity, Basic Award Criteria.
- **Integration:** Can be used to flag items in the graph as "Certified Sustainable" or "Recycled Content." For example, if a user scans a notepad, Fessi can confirm if it is 100% recycled paper based on the Blue Angel GTIN match.[19]

5.3 The "Retailer" Layer: dm-drogerie markt (dmTECH)

- **Source:** dmTECH / dm-drogerie markt website.
- **Role:** A leading German drugstore chain with high data transparency.

- **Relevance:** High for cosmetics, cleaning agents, and organic food—categories often missing from general food databases.
- **Access:** The public developer portal focuses on store data, but product data is structured on their website (e.g., "Material" tabs listing "Flasche aus 100% Recycling-Material").
- **Strategy:** Use as a verification source for top-selling items. If the university has a partnership (e.g., campus store), an API key might be negotiated. Otherwise, data must be manually verified or scraped within legal limits.[20]

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6. Integration Strategy & Data Harmonization

To build Fessi, a robust Extract-Transform-Load (ETL) pipeline must be established to harmonize these disparate sources into a cohesive graph. The graph must act as the translation layer between the user's intent ("Throw away this bottle") and the physical infrastructure.

6.1 The "Material-Stream" Bridge

A core challenge is translating "Material" (Physical) to "Stream" (Logistical). This logic varies by municipality but follows a German standard (The Dual System). The following mapping table illustrates how Fessi must route data:

OFF Tag (packaging_tags)	Material Class	Destination Bin (Standard)	Destination Bin (Berlin/Wertstofftonne)
en:paper, en:cardboard	Paper	Blue Bin (Altpapier)	Blue Bin (Altpapier)
en:glass, de:braunglas	Glass	Glass Container (Brown)	Glass Container (Brown)
en:plastic, en:pet, en:pp	Plastic	Yellow Bin/Sack	Yellow Bin/Wertstofftonne
en:aluminum, en:metal	Metal	Yellow Bin/Sack	Yellow Bin/Wertstofftonne
en:tetra-pak, en:beverage-carto	Composite	Yellow Bin/Sack	Yellow Bin/Wertstofftonne

n			
en:styrofoam (clean)	Plastic	Yellow Bin/Sack	Yellow Bin/Wertstofftonne
en:styrofoam (dirty)	Plastic	Residual Waste	Residual Waste
Unknown/Mixed	Unknown	Residual Waste	Residual Waste

6.2 Handling Composite & Dirty Packaging: Conditional Logic

The Fessi graph must support **conditional logic nodes**. A simple "Product -> Bin" edge is insufficient for items like pizza boxes or unwashed yogurt cups.

- **Scenario:** Pizza Box.
- **Graph Logic:**
  - **Node:** PizzaBox (derived from BSR or OFF).
  - **Edge 1:** hasCondition: Clean $\rightarrow$ DisposeTo: PaperBin.
  - **Edge 2:** hasCondition: Soiled $\rightarrow$ DisposeTo: ResidualBin.
- **Source for Rules:** BSR datasets explicitly list these conditions, which must be modeled as properties on the relationship edges in the graph.[22]

6.3 Implementation Roadmap

1. **Phase 1: Core Ingestion (The "Bread and Butter")**:
   - Download the Open Food Facts MongoDB dump or JSONL export.
   - Filter for countries_tags:en:germany.
   - Extract products with non-null packaging fields.
   - Create WasteItem nodes for the top 5,000 most common products to seed the graph.
2. **Phase 2: Logic Layer (The "Brain")**:
   - Ingest the BSR Abfall-ABC JSON to create the high-level WasteCategory ontology.
   - Link specific OFF products (e.g., "Danone Yogurt") to generic categories ("Yogurt Cup") using simple NLP heuristics or manual mapping.
   - Manually digitize the ZSVR Mindeststandard rules for composite materials and hazardous waste (AVV codes).
3. **Phase 3: Enrichment (The "Nice to Have")**:
   - Import Blue Angel CSV data to add sustainability attributes.
   - Ingest OFF ecoscore data to provide environmental feedback.

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7. Gaps Identified

Despite the richness of the identified sources, gaps remain that the research could not fully resolve:

- **The "Unmapped Tail":** OFF covers retail items well. However, university-specific waste (lab chemicals, bulk cafeteria packaging, imported electronics) is likely missing. This requires a manual entry interface for university staff to add "Local/Campus Items" to the graph.
- **Taxonomy Inconsistencies:** OFF tags are messy (e.g., en:plastic vs en:plastics vs en:plastic-wrapper). A synonym dictionary (SKOS) is required to normalize these into clean Fessi Material Types.
- **Regional Variance:** A user asking "Where does this go?" might be in Munich (Yellow Sack) or on a campus using a private contractor with different rules. The graph must have a Location context node to adjust the query path based on the user's physical location.

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8. Conclusion

Building "Fessi" requires a synthesis of crowdsourced granularity and official regulatory authority. **Open Food Facts** provides the necessary scale for identifying specific grocery items via barcode, offering a "good enough" baseline for packaging materials that can be iteratively improved. This item-level data must be governed by the strict disposal logic derived from **ZSVR** standards and mapped to user-friendly categories provided by municipal sources like **BSR**.

By combining these sources, Fessi can transcend simple "lookup" functionality and offer intelligent, context-aware disposal advice, guiding users not just on *what* an item is, but *how* its specific material composition dictates its fate in the German waste management ecosystem. The proposed hybrid architecture ensures the system is both broad in coverage (via OFF) and accurate in compliance (via ZSVR/BSR), ultimately facilitating higher recycling rates and better waste segregation on campus.

Detailed Source Profiles

Source 1: Open Food Facts (OFF)

- **URL:** https://world.openfoodfacts.org/data
- **Provider:** Open Food Facts (Non-profit)
- **Format:** JSON API, MongoDB Dump, JSONL
- **Access:** Free, Open Data (ODbL)
- **Key Fields:** code (GTIN), packaging_tags, ecoscore_grade
- **Relevance: Essential (5/5)**

Source 2: ZSVR Mindeststandard

- **URL:** https://www.verpackungsregister.org
- **Provider:** Stiftung Zentrale Stelle Verpackungsregister
- **Format:** PDF Documents
- **Access:** Free Download
- **Key Content:** Legal definitions of recyclability, composite thresholds
- **Relevance: Essential (5/5)** (Logic Layer)

Source 3: BSR Abfall-ABC

- **URL:** https://daten.berlin.de/datensaetze/bsr-recyclingh-fe-und-abfall-abc
- **Provider:** Berliner Stadtreinigung
- **Format:** JSON, CSV
- **Access:** Free, CC BY License
- **Key Fields:** Item Name, Disposal Route, Avoidance Tips
- **Relevance: High (4/5)** (Ontology/Mapping)

Source 4: Blue Angel E-Commerce Data

- **URL:** https://www.blauer-engel.de/en/trade/e-commerce
- **Provider:** RAL gGmbH / German Government
- **Format:** CSV/JSON (via registration)
- **Access:** Free (Registration Required)
- **Key Fields:** GTIN, Certification Status
- **Relevance: Medium (3/5)** (Enrichment)

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